III. State whether true or false. If false, correct the statement.

1. Electrically neutral means it is either zero or equal positive and negative charges.
2. Ammeter is connected in parallel in any electric circuit.
3. The anode in electrolyte is negative.
4. Current can produce magnetic field.

IV. Fill in the blanks.

1. Electrons move from _____ potential to _____ potential.
2. The direction opposite to the movement of electron is called _____ current.
3. The e.m.f of a cell is analogues to _____ of a pipe line.
4. The domestic electricity in India is an ac with a frequency of _____ Hz.

V. Conceptual questions.

1. A bird sitting on a high power electric line is still safe. How?
2. Does a solar cell always maintain the potential across its terminals constant? Discuss.
3. Can electroplating be possible with alternating current?

VI. Answer the following.

1. On what factors does the electrostatic force between two charges depend?
2. What are electric lines of force?
3. Define electric field.
4. Define electric current and give its unit.
5. State Ohm's law.
6. Name any two appliances which work under the principle of heating effect of current.
7. How are the home appliances connected in general, in series or parallel. Give reasons.
8. List the safety features while handling electricity.

VII. Exercises.

1. Rubbing a comb on hair makes the comb get – 0.4C. (a) Find which material has lost electron and which one gained it. (b) Find how many electrons are transferred in this process.
2. Calculate the amount of charge that would flow in 2 hours through an element of an electric bulb drawing a current of 2.5 A.
3. The values of current (I) flowing through a resistor for various potential differences V across the resistor are given below. What is the value of resistor?

I (ampere)	0.5	1.0	2.0	3.0	4.0
V (volt)	1.6	3.4	6.7	10.2	13.2

[Hint: plot V-I a graph and take slope]

REFERENCE BOOKS

Fundamentals of Physics - K.L Gomber andK. L.Gogia

Concepts of Physics - H.C Verma

General Physics - W.L. Whiteley

INTERNET RESOURCES

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https://www.explainthatstuff.com/batteries.html

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